LAB 3 — COMPONENT MODELLING & ARCHITECTURAL PATTERN

Student Club Event Ticketing & Budget Portal — Written Justification

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Architecture	Layered (3-tier) Architecture

Architecture Selection: We chose Layered (3-tier) Architecture for the Student Club Event Ticketing & Budget Portal.

Reason 1 — Separation of responsibilities: The Presentation Layer contains the Web Portal UI for Club Leads, approvers, administrators, and students. The Business/Application Layer contains the Auth & RBAC Service, Proposal & Budget Manager, Approval Workflow Engine, QR Ticketing Service, Notification Service, and Dashboard & Reporting. The Data Layer centralizes repository and database access. Separating these responsibilities makes the portal easier to maintain, test, and extend.

Reason 2 — Fits the approval and ticketing workflow: The portal supports a clear sequence: a Club Lead submits an event proposal and budget, role-based access is checked, the proposal moves through the approval workflow, and final approval triggers QR ticket generation. Notifications keep users informed, while the dashboard presents role-filtered proposal and budget reports. A layered design organizes this flow without requiring every function to be deployed as a separate microservice.

Security advantage: The Auth & RBAC Service checks user sessions and permissions before protected actions are allowed. Role-based access helps ensure that Club Leads, approvers, administrators, and students access only the functions appropriate to their roles. Persistent data access is routed through the Data Access Layer, centralizing database operations.

Performance benefit: The dedicated Data Access Layer allows repository logic and SQL access to be managed consistently. Keeping the business components separated makes it easier to identify and optimize slow operations. The notification and email gateway and QR ticketing functions are separated from proposal management, reducing coupling between responsibilities.

Components and interfaces: The diagram includes the Web Portal UI, Auth & RBAC Service, Proposal & Budget Manager, Approval Workflow Engine, QR Ticketing Service, Notification Service, Dashboard & Reporting, Data Access Layer, and Portal Database, plus an external Email/SMTP Gateway. Interfaces cover authentication, proposal submission, approval, ticket booking/download, reporting, authorization, workflow routing, QR ticket generation, notifications, repository access, and SQL queries. Provided interfaces are shown as balls and required interfaces as sockets.

Note: This is a proposed design for the Student Club Event Ticketing & Budget Portal.